Program Design Quiz 7

(Please use C language to solve the problem)

題目敘述：

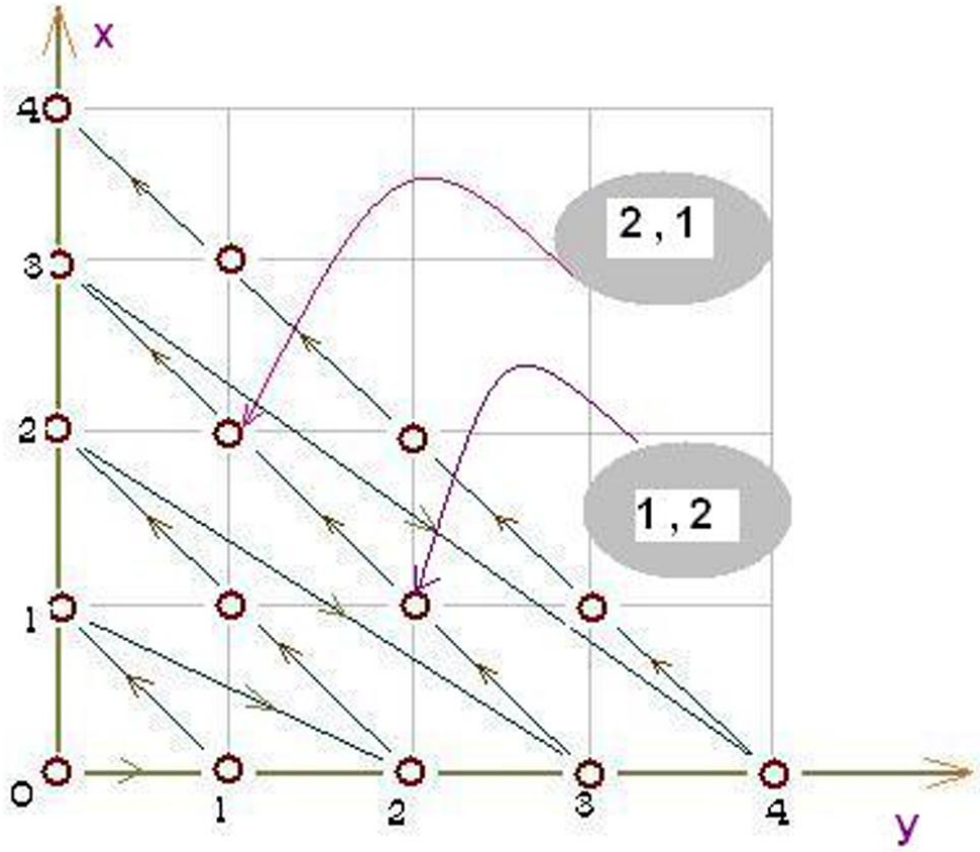
如下圖所示，在平面直角座標系上，一些圓點標示在不同的整數座標位置。圖中還畫有許多**有方向的箭頭**，表示你可以沿著箭頭指向的方向，從一個圓點走到另一個圓點；**不能**逆著箭頭方向走回去。

我們定義「從一個圓點走到另一個圓點所需的步數」為：在路徑上**經過的中介圓點個數 + 1**

例如，要從座標 $(0,3)$ 走到 $(3,0)$，你會依序經過 $(1,2)$ 和 $(2,1)$ 這兩個中間的圓點，所以所需步數為 $2+1=3$。

在本題中，請你從已知的起點與終點座標，計算**若沿著圖中的箭頭前進，從起點走到終點共需要多少步**。

你可以假設，對於輸入中的每一筆資料，**從起點一定可以走到終點**，也就是說，合法路徑必定存在。



輸入說明：

輸入的第一行包含一個整數 n ($0 < n \leq 500$)，代表測試資料的筆數。

接下來共有 n 行，每一行包含四個整數 x1 y1 x2 y2，其整數介於 0~100000 之間。其中：

- (x1, y1) 為起點圓的座標
- (x2, y2) 為終點圓的座標

所有座標皆以 (x, y) 形式表示。

輸出說明：

對於每一筆測試資料，請輸出一行，格式為：

Case k: ans

其中：

- k 為測試資料編號（從 1 開始遞增）
- ans 為從起點圓 (x1, y1) 走到終點圓 (x2, y2) 所需的步數

Examples

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Input 1:

3

0 0 0 1

0 0 1 0

0 0 0 2

Output 1:

Case 1: 1

Case 2: 2

Case 3: 3

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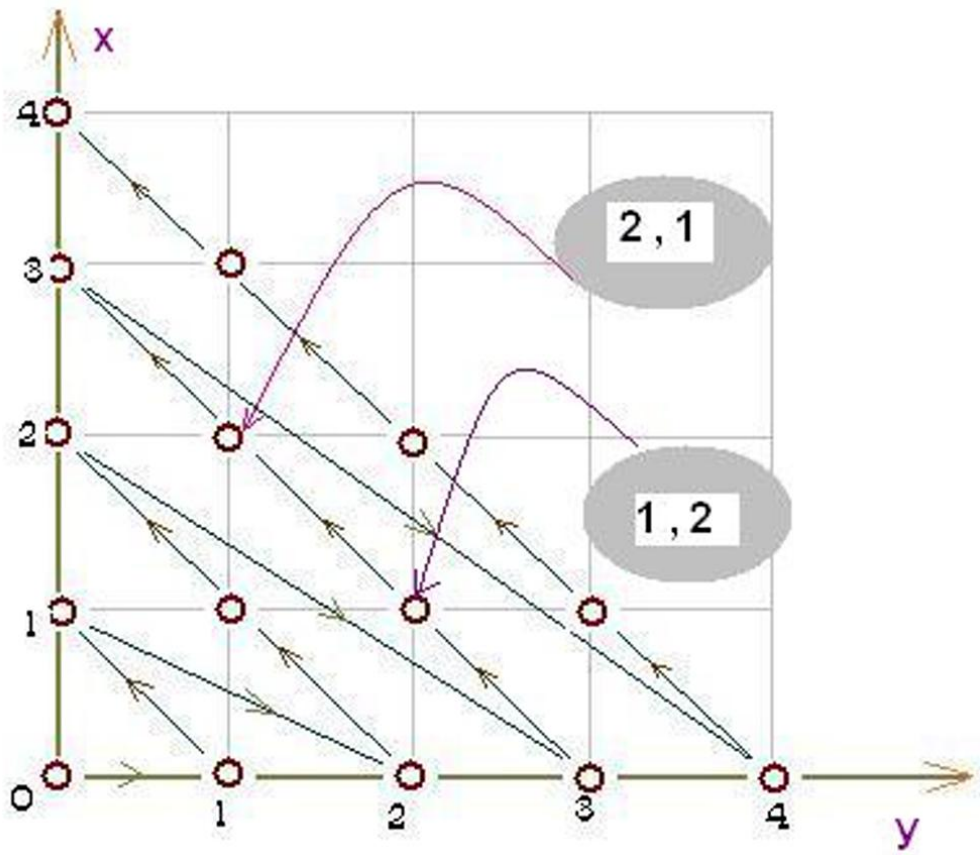
Problem Description：

First take a look at the following picture. In this picture, each circle has a coordinate according to Cartesian Coordinate System. You can move from one circle to another following the path denoted by forward arrow symbols. To go from a source circle to a destination circle,

total_number_of_step(s)_needed = number_of_intermediate_circles_you_pass + 1

For example, to go from (0,3) to (3,0) you have to pass two intermediate circles (1,2) and (2,1). So, in this case, total number of steps needed is 2+1 = 3. In this problem, you are to calculate number of step(s) needed for a given source circle and a destination circle. You can assume that, it is not possible to go back using the reverse direction of the arrows.



Input：

The first line of the input contains an integer $n(0 < n \leq 500)$, which denotes the number of test cases.

Each of the following $n$lines contains four integers x_1, y_1, x_2, y_2, each in the range from 0 to 100000 (inclusive). Here:

- $(x_1, y_1)$is the coordinate of the starting circle.
- $(x_2, y_2)$is the coordinate of the destination circle.

All coordinates are given in the form (x, y).

Output：

For each test case, output one line in the following format:

Case k: ans

where

- $k$is the test case number (starting from 1).
- $ans$is the number of steps required to go from the starting circle $(x_1, y_1)$to the destination circle (x_2, y_2).

Examples

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Input 1:

3

0 0 0 1

0 0 1 0

0 0 0 2

Output 1:

Case 1: 1

Case 2: 2

Case 3: 3